

Developing an App to Investigate Whether an Automatically Adjusting Active Filter can Mitigate CVD

Background

Color Vision Deficiency (CVD), also known as colorblindness, glasses have become widely available, however, their one size fits all approach limits their effectiveness. This coupled with a high price tag limits their accessibility. This study explores a novel design for an app with an automatically adjusting CVD filter with an integrated mosaic CVD test.

Color Vision Deficiency (CVD) occurs when someone is missing one of three color detecting cones in their eyes. The amount of cones of each color missing determines the strength of color blindness that the individual has.

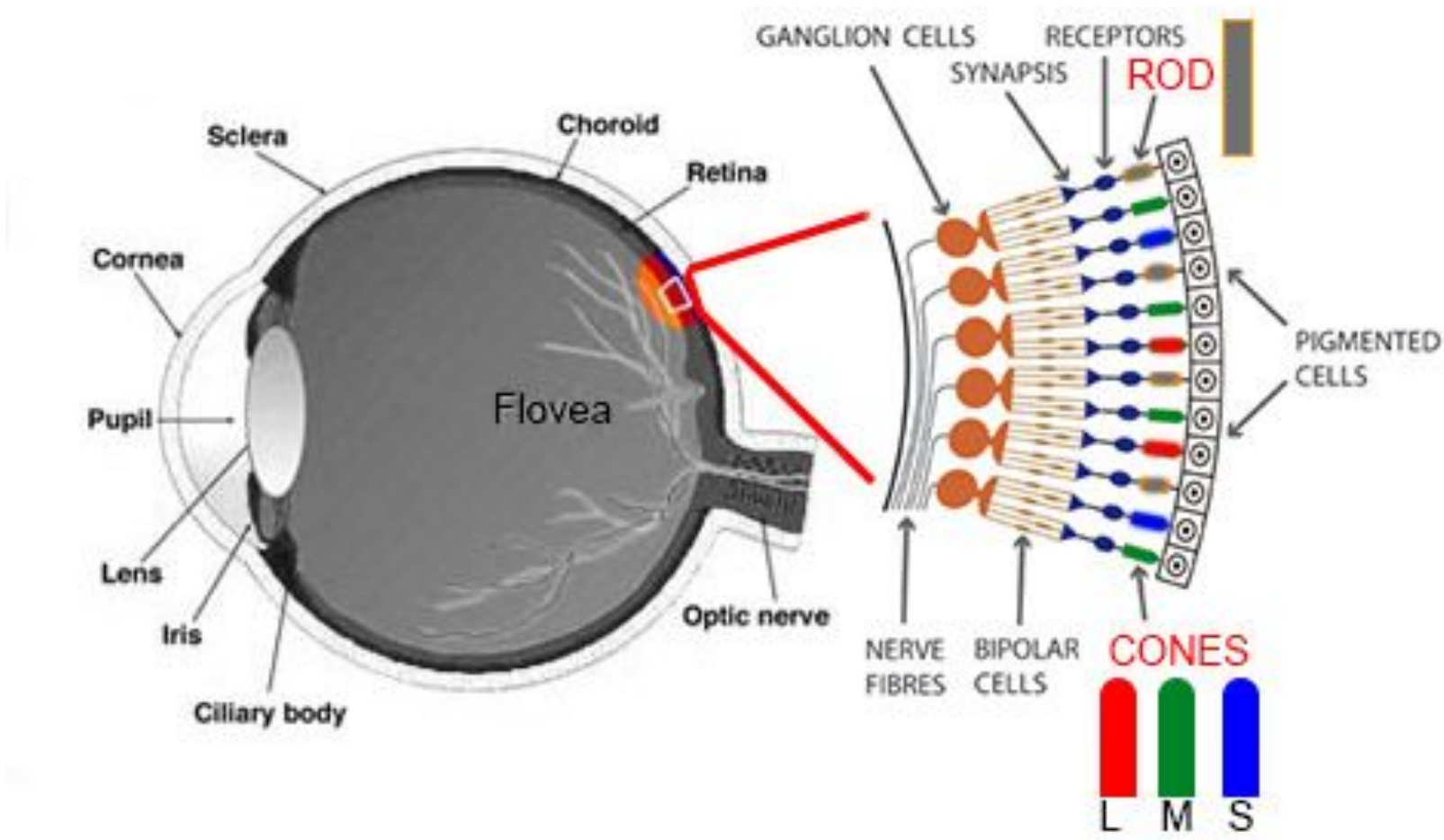


Figure 2: Color detecting cones in human eyes (Syntax Terror, n.d)

CVD affects roughly 4.25% or 344 million people world wide. It can prevent people from getting jobs in the military, medicine, science, engineering, and aviation.

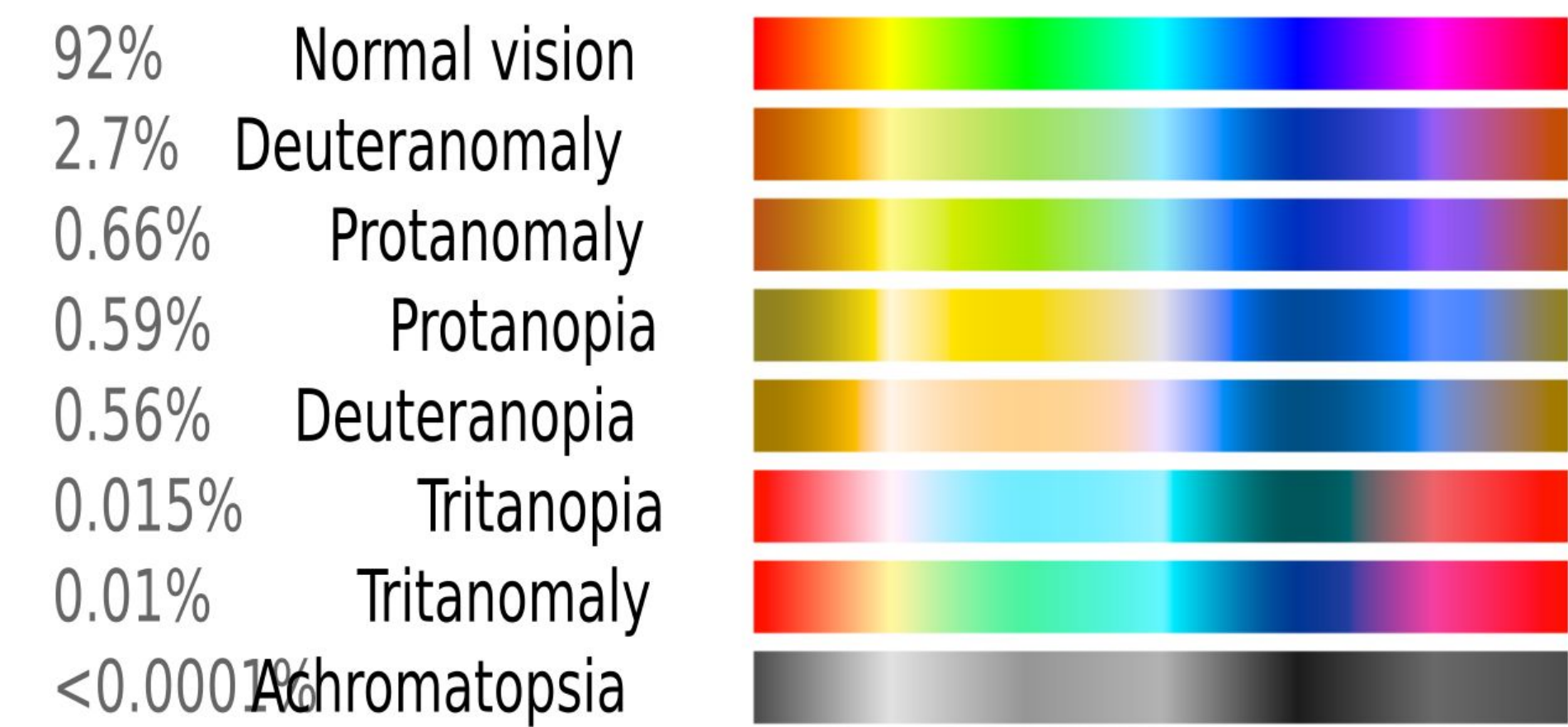


Figure 3: All types of CVD, and what percentage of CVD they make up. (Lenstore, February 13, 2019)

Physical lenses such as glasses or contact lenses are not as effective as digital lenses due to their inability to be easily adjusted according to the wearers needs. As such, digital lenses have the potential to provide much more utility to colorblind individuals.

Additionally, many physical lenses are made out of materials such as 3d printer resin, which may not be completely safe to use as glasses or contact lenses.

Hypothesis:
We hypothesize that an automatically adjusting active filter can mitigate the effects of red-green CVD without requiring human input to change the filter strength.

Introduction and Method

Independent variable: Mosaic test CVD results

Dependent variable: The score obtained on the mosaic CVD test.

Constants: The amount of plates in the test, the type of CVD each plate tests for, and the difficulty of each plate.

Control: Test subject

Materials: Computer, phone, and a test subject

Intended Procedure:

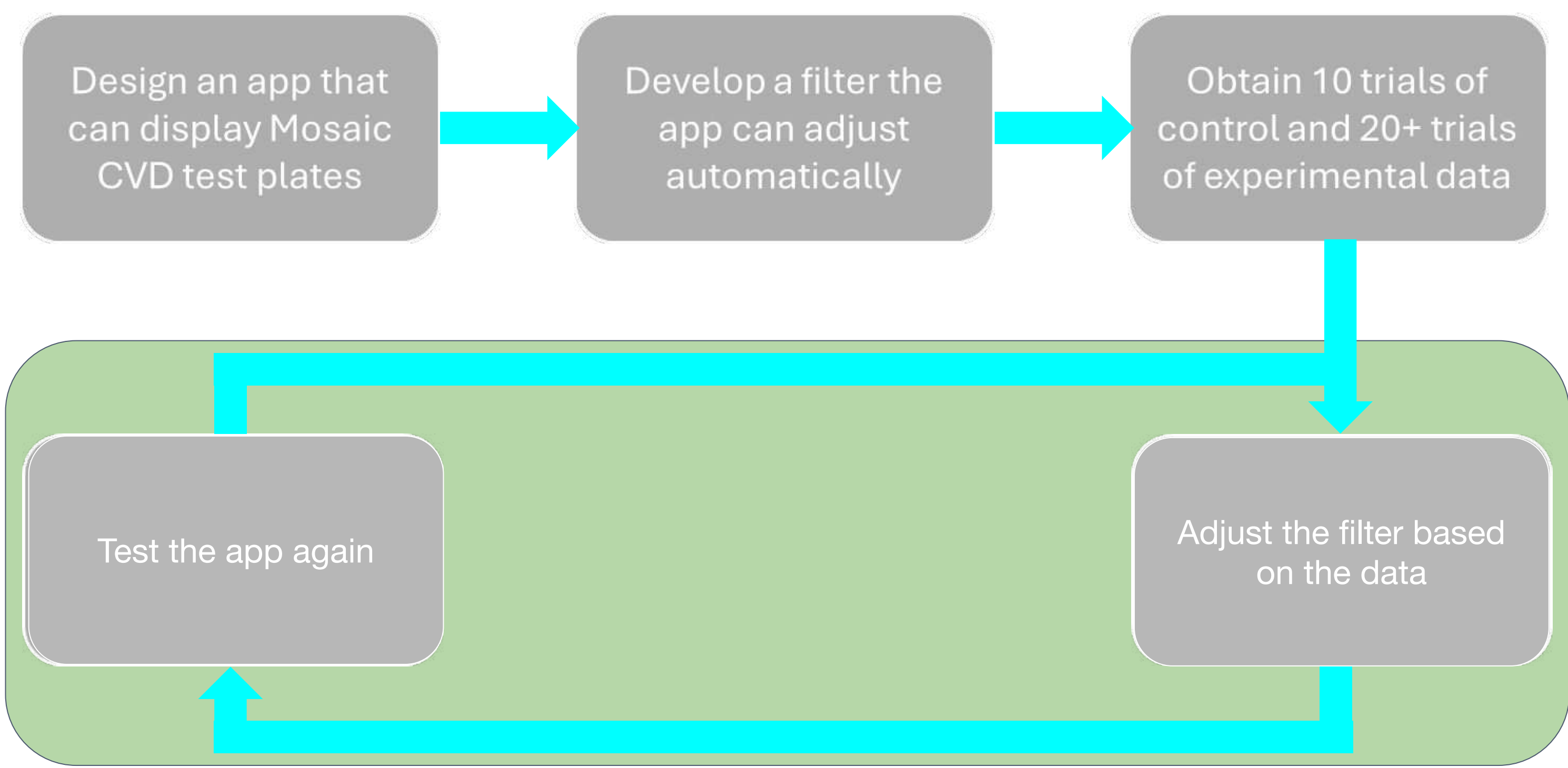


Fig 4: Screenshot of a mosaic CVD test plate in the app

Data Analysis and Results

The data That we have collected so far has demonstrated that the active filter and testing systems implemented in the app are able to effectively mitigate the effects of red-green colorblindness. The original score without a filter is 80.60%, but with the filter it jumps to 96.27%, an improvement of 15.47%, which is very significant.

Conclusion

Based off of the fact that the mosaic test scores improved by an average of 15.47%, it is safe to say that the app works as intended. In its current state the app is ready to be published for public access, the only hurdle would be finding an appropriate platform. Alternatively, the web access could just be maintained for the app to be used through a web browser, which would still work, but would nonetheless be less optimal than an app.

Additionally, the app could be adjusted to mitigate other forms of colorblindness. The only things that would require adjustment would be the colors tested in the mosaic test and the color of the filter. The only reason why this was not tested during this experiment was the combination of a lack of time and the lack of proper test participants.

Citations/Acknowledgements

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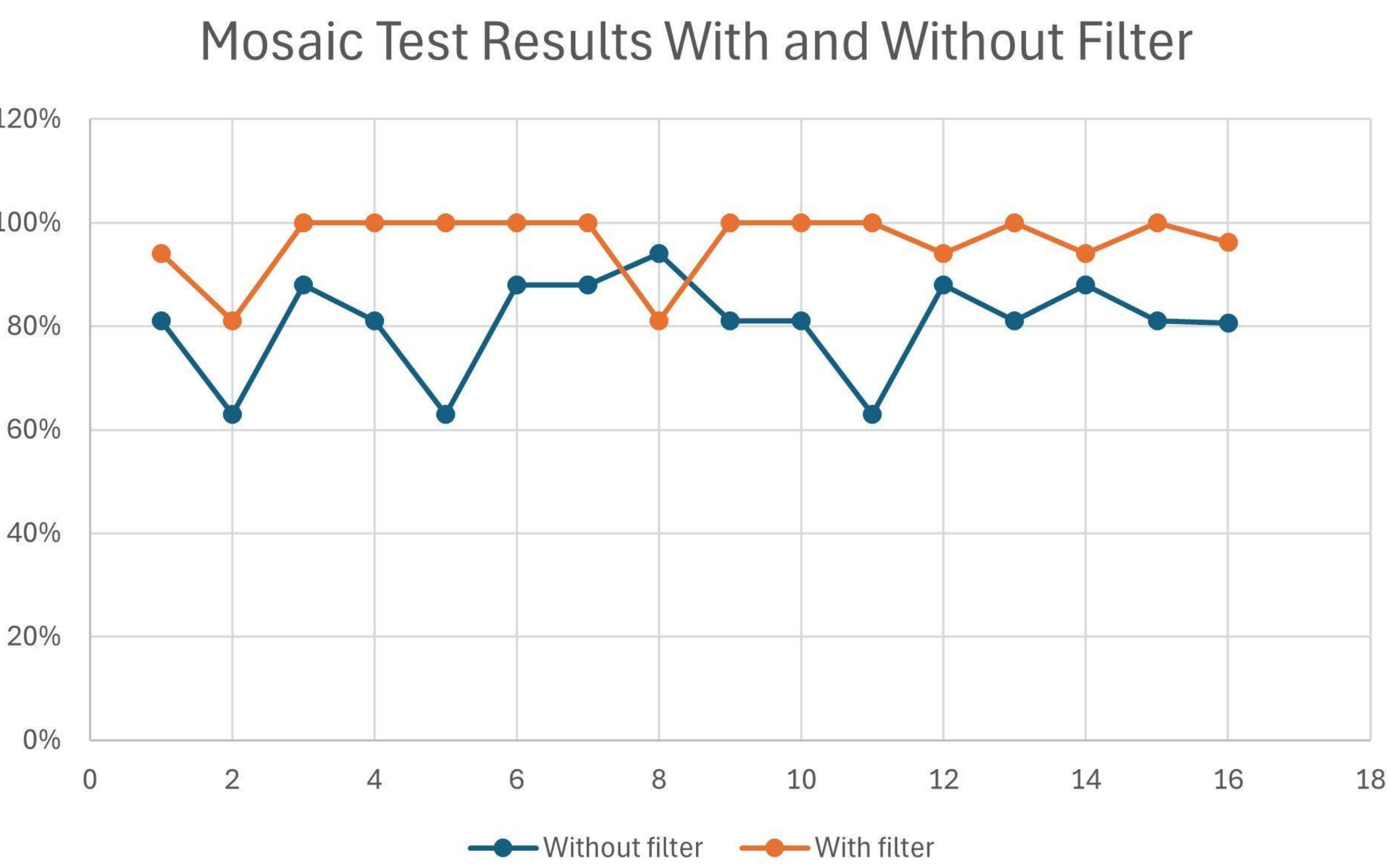
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Data



	Without filter	With filter
Trial 1	81%	94%
Trial 2	63%	81%
Trial 3	88%	100%
Trial 4	81%	100%
Trial 5	63%	100%
Trial 6	88%	100%
Trial 7	88%	100%
Trial 8	94%	81%
Trial 9	81%	100%
Trial 10	81%	100%
Trial 11	63%	100%
Trial 12	88%	94%
Trial 13	81%	100%
Trial 14	88%	94%
Trial 15	81%	100%
Average	80.60%	96.27%